

BİLGE ZERDA KEKLİK

SOFTWARE ENGINEER · DATA & ML · SOFTWARE DEVELOPMENT

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SUMMARY

Software Engineering graduate with hands-on experience in full-stack development, data/ML projects, and algorithmic problem solving. Experienced with Python, FastAPI, React, PostgreSQL, scikit-learn, and Git, with project experience spanning natural-language processing, recommendation systems, classification, and route optimization. Interested in data-driven software engineering, machine learning, and applied R&D.

EDUCATION

Turkish Aeronautical Association University — Faculty of Engineering

B.Sc. in Software Engineering · English-taught program · 2023 – 2026

EXPERIENCE

Academic Assistant — Part-Time | Software Engineering

Turkish Aeronautical Association University, Faculty of Engineering · Mar 2025 – Jun 2026 · 3 academic terms · 12 months total

- Supported Software Engineering research assistants with laboratory sessions, exam proctoring, documentation, data organization, and departmental tasks.
- Selected for a 5-member TTO project team and contributed as a frontend developer using React.
- Developed student/university project management, authentication, membership, and address-book interfaces.
- Worked with Git/GitHub throughout the project lifecycle and collaborated through regular Microsoft Teams meetings.
- Initially prepared for backend development before transitioning to frontend based on team needs; completed the assigned frontend scope ahead of schedule.

SELECTED PROJECTS

TravelMind — AI-Powered Personalized Travel Planning System · [GitHub](#)

Python · FastAPI · React · PostgreSQL · SQLAlchemy · scikit-learn · Google Maps API · JWT

- Developed the full-stack application, implementing the FastAPI backend, PostgreSQL/SQLAlchemy data layer, REST APIs, JWT authentication, and React frontend.
- Integrated and further developed the team's Turkish NLP/recommendation component, connecting preference processing and recommendation workflows to the application.
- Curated and expanded the project's 65-activity Antalya dataset, including activity information and visual data, to support the recommendation system.
- Implemented Dijkstra-based route optimization using Haversine distances and integrated Google Maps for route visualization and itinerary planning.

ML Classification Toolkit — Interactive Classification & Evaluation · [GitHub](#)

Python · scikit-learn · pandas · NumPy · Streamlit

- Developed an interactive Streamlit toolkit for CSV-based preprocessing, classification, model training, and evaluation.
- Implemented Perceptron, MLP, and Decision Tree classification workflows with configurable MLP architectures.
- Built preprocessing and evaluation workflows covering scaling, encoding, accuracy, precision, recall, F1-score, and confusion matrices.

CPU Scheduling Simulator — Operating Systems Project · [GitHub](#)

C++ · Python · JavaScript

- Implemented FCFS, SJF, Round Robin, and Priority Scheduling algorithms with scheduling metrics.
- Reimplemented the algorithms as an interactive browser-based simulator featuring Gantt charts and algorithm comparison.

TECHNICAL SKILLS

Languages: Python · C++ · JavaScript · SQL

Backend & Database: FastAPI · REST APIs · PostgreSQL · SQLAlchemy · JWT

Data & ML: scikit-learn · pandas · NumPy · Streamlit

Frontend: React · HTML/CSS

Tools: Git · GitHub · Docker

CERTIFICATIONS & SELECTED PROGRAMS

• **Ministry of Industry and Technology — National Technology Initiative**
Autonomous Driving Technologies Training Program · Certificate Earned ·
2026 Certificate No. 40412928295801 · [Verify Certificate](#)

• **Amazon Web Services Training and Certification**
AWS Academy Graduate · Cloud Foundations · Training Badge ·
2026 · [Verify Badge](#)

• **TEI Aviation Engines School**
5th Aviation Engines School Training Program
Certificate of Achievement · 2025